Last year, generative AI took the world by storm, as users discovered tech that could churn out anything from poetry to exotic imagery just with a text prompt. The generative AI market is expected to exceed $100 billion by 2030, growing at an average rate of more than 35 per cent per year. With the advances in generative AI over the past year, detecting deepfakes has become an even bigger challenge. According to Professor Amit Roy-Chowdhury, who leads the Video Computing Group (VCG) at the University of California, Riverside, US, identifying whether an image is real or synthetic is largely becoming a matter of context. "In the sense that you look at an image and then someone says 'no, this never happened'." This, incidentally, is what happened in both the Rashmika and Katrina deepfakes.

HOW TO SPOT A DEEPFAKE

Roy-Chowdhury's research work in computer vision and machine learning has included methods to identify deepfakes. "At the end of the day, on a computer, an image is a matrix of numbers and, if you look at that set of numbers, they have certain characteristics in them. In a natural image, numbers occur in a certain way." Hence, when an image is manipulated, its statistics change too. For example, if someone's hair has been digitally altered, the pixels from that region could provide the clues—when one pixel of hair has one value, but the neighbouring one has a very different one even though they are both the same colour. "If that happens over a number of pixels, you can start suspecting why it is happening. A person's hair does not change like that naturally," he explains.

In 2022, VCG developed a new method that it claims can detect manipulated facial expressions in deepfake videos with higher accuracy than current state-of-the-art methods. Detecting manipulations in a person's facial expressions is arguably a more difficult job than spotting face swaps. "Expression-changes are much more subtle. You have to look at very small local regions in an image to see if there have been manipulations." VCG's efforts bring researchers a step closer to developing automated tools to spot manipulated videos that contain propaganda or misinformation.

India saw an example of facial expression manipulation in May 2023, bang in the middle of a pivotal moment in the five-month-long protest by female Olympic wrestlers pressing for investigation into complaints of sexual harassment. When the Delhi Police detained the protesters, wrestlers Vinesh and Sangeeta Phogat posted a selfie of themselves sitting grimly in a police van with the cops. But it was another version—of both women smiling at the camera—that circulated virally until it was debunked as a fake. "Any person of importance, and not just in Bollywood, is suddenly petrified with deepfakes, which have been around but have been amplified with the power of AI," says Shrenik Gandhi, co-founder and CEO, White Rivers Media, a digital marketing agency that works with leading Bollywood studios. "The biggest issue is that negativity travels much faster than positivity. If anything negative goes out, even

RECENT DEEPFAKES IN INDIA



KATRINA KAIF

The original image from the film *Tiger 3* has been morphed to show the actress in different clothes



IF DEEPFAKES ARE A PRODUCT OF NEW TECH, THE SOCIAL NETWORKS ON WHICH THEY TRAVEL AND BECOME VIRAL IS THE OTHER ASPECT OF THE EQUATION